

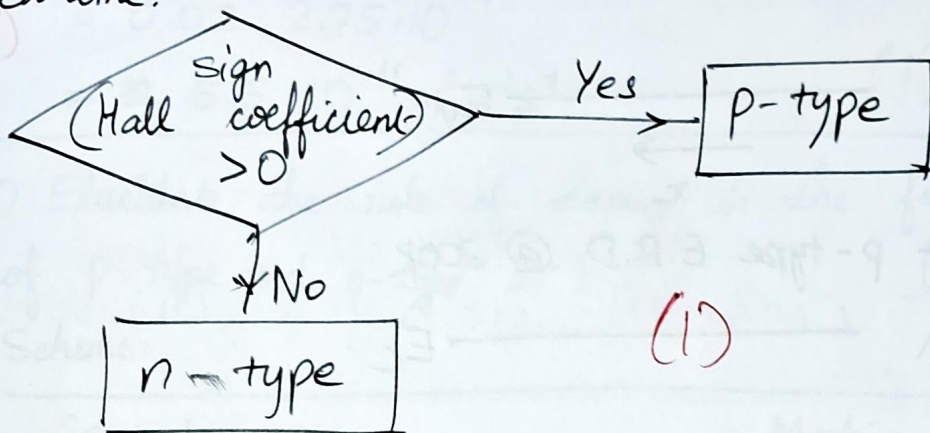
Sat, 24th January, 2026

Summative Assessment with scheme of evaluation

Section - A

①(a) How to distinguish the n-type and p-type semiconductor using Hall effect? [1M]

Scheme:



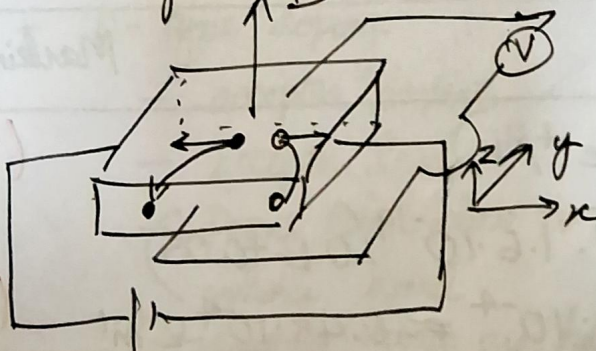
①(b) Evaluate the Hall voltage on s.c. with application of $\vec{E}$ and $\vec{B}$ mutually perpendicular to each other. [4M]

Scheme:

Concept

Marking

— Diagram $\vec{B}$



— Lorentz force expression

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

(1)

— Equilibrium condition

$$\vec{F}_{\text{net}} = 0$$

$$E_H = vB$$

(1)

— Derivation of Hall field/voltage

$$V_H = E_H \cdot y$$

$$= v y B$$

$$j = nev = \frac{I}{yt}$$

$$\Rightarrow V_H = \frac{IB}{net}$$

(1)

①(c) The electrical conductivity of n-type semiconductor is $0.02 \Omega^{-1}m^{-1}$. The s.c. is carrying current of 20A. is placed in magnetic field of strength 0.4T, which is having a thickness of 1mm produces a Hall voltage of $22\mu V$ across its faces. Calculate the mobility of charge carriers. [3M]

Scheme:

Concept

Marking

— Hall coefficient

$$R_H = \frac{V_H t}{I B}$$

$$= \frac{22 \cdot 10^{-6} \cdot 10^{-3}}{20 \cdot 0.4}$$

$$= 2.75 \cdot 10^{-9} \text{ m}^3/\text{C}$$

(2)

— Mobility

$$\mu = \sigma R_H$$

$$= 0.02 \cdot 2.75 \cdot 10^{-9}$$

$$= 5.5 \cdot 10^{-11} \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$$

(1)

② (a) Elucidate the role of doping in the formation of p-type and n-type s.c.'s. [4M]

Scheme:

Concept	Marking
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— n-type doping

- donor dopants.
- donor level
- Fermi level close to conduction band.
- majority carriers are electrons

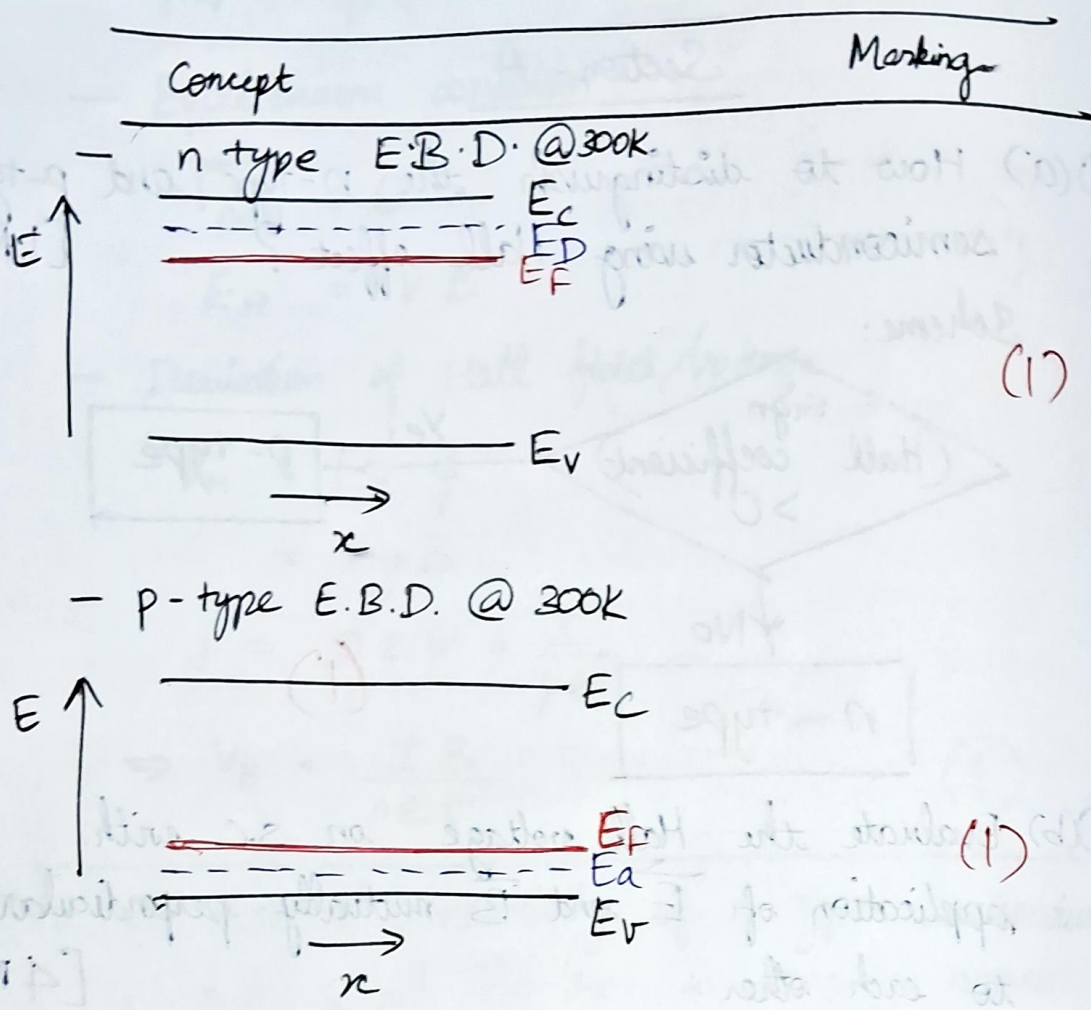
(2)

— p-type doping

- acceptor dopants
- acceptor level
- Fermi level close to valence band
- majority carriers are holes

(2)

②(b) Draw the Fermi levels of n-type and p-type s.c.'s. [2M]



②(c) Intrinsic carrier concentration of Si is $1.5 \cdot 10^{15} \text{ m}^{-3}$ and mobilities of electrons and holes are $0.13 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $0.05 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$. Calculate electrical conductivity. [2M]

Scheme:

Concept	Markings
$\sigma_i = n_i e (\mu_e + \mu_h)$	(1)
$= 1.5 \cdot 10^{15} \cdot 1.6 \cdot 10^{-19} (0.13 + 0.05)$	(1)
$= 2.4 \cdot 0.18 \cdot 10^{-4} \approx 0.48 \cdot 10^{-4} \text{ } \Omega^{-1} \text{ m}^{-1}$	

③(a) Write the postulates of QFET. [3M]

Scheme:

Concept

Marking

- Matter wave postulate
- Pauli exclusion principle
- Fermi-Dirac statistics

(1)

(1)

(1)

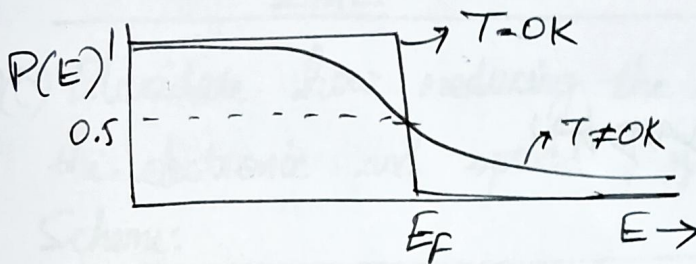
③(b) Sketch two Fermi-Dirac curves on the same graph for two different temperatures and explain the difference

Scheme:

[2M]

Concept

Marking



(1)

(1)

③(b) Explain the variation of f-D distribution function at $T=0K$ and $T \neq 0K$ [3M]

Scheme:

Concept

- $T=0K$

$$\begin{aligned} P(E < E_f) &= \frac{1}{1 + \exp\left(\frac{E - E_f}{0}\right)} \\ &= \frac{1}{1 + \exp(-\infty)} \\ &= 1 \end{aligned}$$

(1)

$$P(E > E_F) = \frac{1}{1 + \exp\left(\frac{E - E_F}{0}\right)}$$

$$= \frac{1}{1 + \exp(\infty)}$$

$$= 0$$

(1)

- $T \neq 0K$

(1)

$$P(E = E_F) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

$$= \frac{1}{1 + \exp(0)}$$

$$= \frac{1}{2}$$

④ (a) Elucidate the concept of energy quantization. [1M]

Scheme:

Concept

Marking

$$E = h\nu$$

(1)

energy is quantized

④ (b) Obtain the expression for the energy of a particle in a one-dimensional potential well [4M]

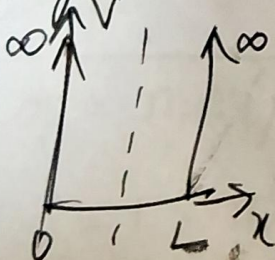
Scheme:

Concept

Marking

- Diagram

(1)



- Schrodinger equation

(1)

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + E\psi = 0$$

- Boundary condition

(1)

$$\psi = A \sin kx + B \cos kx$$

$$\psi(0) = 0 \Leftrightarrow B = 0$$

$$\psi(L) = 0 \Leftrightarrow \sin kL = 0$$

- Energy levels

(1)

$$k = \frac{n\pi}{L}$$

$$E = \frac{\hbar^2 k^2}{2m} \\ = \frac{n^2 \hbar^2 \pi^2}{2mL^2}$$

④(c) Elucidate how reducing the size of object affects the electronic and optical properties in quantum dots.

Scheme:

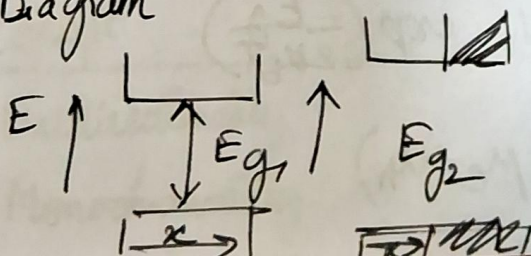
Concept

Marking

- Size $\propto \frac{1}{\text{Bandgap}}$

- ~~Diagram~~ Bandgap $\uparrow$, Conductivity $\downarrow$ (1)
Bandgap $\uparrow$, λ_{photon} $\downarrow$ (1)

- Diagram



(1)

Section - B

- ⑤ (a) How does a light emitting diode work?
Explain why light is produced when an electric current passes through an LED? [5M]

Concept

Markings

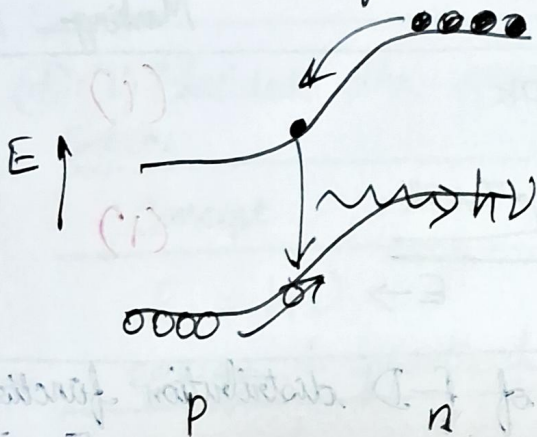
- Principle

Electroluminescence

(1)

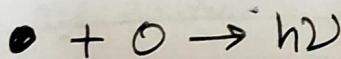
- Band diagram

(1)



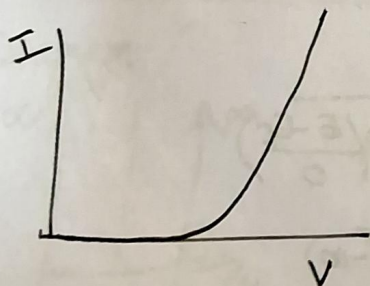
- Recombination mechanism

(1)



- I-V characteristics

(1)



- Explanation

(1)

⑤(b) Obtain the relation between bandgap and wavelength of the emitted colour in LED. [2M]

Concept	Marking
$E_C - E_V = h\nu$	(1)
$E_g = \frac{hc}{\lambda}$	(1)

⑤(c) Write the essential conditions to obtain LASER. [3M]

Concept	Marking
- Conditions	
High photon density	(1)
Large lifetime of excited state	(1)
High population of excited state	(1)
- Components	
Active medium	
Pump	
Optical resonator	

⑤(d) Both LASERS and LEDs convert the electrical energy into light energy. How they are different from each other? Explain. [4M]

Concept	Marking
Unidirectionality	(1)
Monochromaticity	(1)

Coherence

(1)

High intensity

(1)

Low pulse width

Q(a) How electrical conductivity varies with increasing the temperature in semiconductors and conductors. [3M]

Concept	Marking
$\sigma = \frac{ne^2\tau}{m}$	(1)
— Conductors As $T \uparrow$, $\tau \downarrow$ due to more collisions so $\sigma \downarrow$	(1)
— Semiconductors As $T \uparrow$, $n \uparrow\uparrow$. Though $\tau \downarrow$, increase in n dominates decrease in τ $\therefore \sigma \uparrow$	(2)

Q(b) Obtain an expression for the electrical conductivity of an intrinsic semiconductor from the expressions of electrons and holes. [4M]

Concept	Marking
$n_i = \sqrt{N_c N_v} \exp\left(\frac{-E_g}{2k_B T}\right)$	(2)
$\sigma_i = n_i e (\mu_e + \mu_h)$	(1)

$$\sigma_i = \sigma_0 \exp\left(-\frac{E_g}{2k_B T}\right) \quad (1)$$

$$\text{where } \sigma_0 = \sqrt{N_c N_v} e (\mu_e + \mu_h)$$

6(c) The bandgap of Ge and Si are 0.7 eV and 1.1 eV. The electrical conductivity of semiconductor follows the relation $\sigma = K \exp\left(-\frac{E_g}{2k_B T}\right)$. Calculate the temperature at which conductivity of Si is same as that of Ge at room temperature. [3M]

Concept

Marking

$$\sigma_0 \exp\left(-\frac{E_{g1}}{2k_B T_1}\right) = \sigma_0 \exp\left(-\frac{E_{g2}}{2k_B T_2}\right) \quad (1)$$

$$\frac{E_{g1}}{2k_B T_1} = \frac{E_{g2}}{2k_B T_2} \quad (1)$$

$$\frac{1.1}{T_1} = \frac{0.7}{300}$$

$$T_1 = \frac{300 \times 1.1}{0.7} \quad (1)$$

$$= 471.4 \text{ K}$$

6(d) σ of Ge at 20°C is $2 \Omega^{-1} \text{ m}^{-1}$. $E_g = 0.7 \text{ eV}$. Calculate σ at 40°C [4M]

Concept

Marking

$$T_1 = 20 + 273 = 293 \text{ K} \quad (1)$$

$$T_2 = 40 + 273 = 313 \text{ K}$$

$$\sigma = \sigma_0 \exp\left(\frac{-E_g}{2k_B T}\right)$$

$$\sigma_1 = \sigma_0 \exp\left(\frac{-E_g}{2k_B T_1}\right)$$

$$\sigma_2 = \sigma_0 \exp\left(\frac{-E_g}{2k_B T_2}\right)$$

$$\frac{\sigma_1}{\sigma_2} = \frac{\exp\left(\frac{-E_g}{2k_B T_1}\right)}{\exp\left(\frac{-E_g}{2k_B T_2}\right)}$$

(1)

$$\frac{\sigma_1}{\sigma_2} = \exp\left(\frac{-E_g}{2k_B} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right)$$

$$\ln \sigma_1 - \ln \sigma_2 = \frac{-E_g}{2k_B} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\begin{aligned} \text{Now RHS} &= \frac{-0.7 \cdot 1.602 \cdot 10^{-19}}{2 \cdot 1.38 \cdot 10^{-23}} \left(\frac{1}{293} - \frac{1}{313}\right) \\ &= \frac{-0.7 \cdot 1.602 \cdot 10^{-19}}{2 \cdot 1.38} (0.003412 - 0.003194) \\ &= -8.82 \cdot 10^{-7} \end{aligned}$$

(1)

$$\ln \frac{\sigma_1}{\sigma_2} = -0.882$$

$$\Rightarrow \frac{\sigma_1}{\sigma_2} = \exp(-0.882) = 0.413$$

(1)

$$\Rightarrow \sigma_2 = \frac{\sigma_1}{0.413} = 2.45 \cdot \sigma_1 = 4.83 \text{ cm}^{-1}$$

END OF PAPER